1. Product Overview

1.1 Problem Statement

University students, corporate workers, and NYSC members frequently face poor spending discipline, resulting in overspending early in the month and a lack of structured daily financial control.

1.2 Solution

SplitSpend is a fintech platform that enforces financial discipline through daily spending budgets. Key capabilities include:

- Lock deposited funds into auto-calculated daily spending limits
- Allow parents to remotely manage and cap their child's daily university spending
- Enable zero-fee vendor payments within the platform — purely internal wallet moves
- Allow users to pay other SplitSpend users directly — no external payment processor involved
- Provide a Main Balance fallback to avoid frustrating restrictions
- Allow users to transfer funds from Main Balance to any Nigerian bank account

1.3 Unique Value Proposition

Differentiator	Details
Forced Discipline	Money is locked into daily budgets — not freely accessible
Parent Controls	Parents can fund and set daily limits for students remotely
Zero-Fee Payments	Vendor and user-to-user payments are internal wallet moves — instant, free, no Paystack round-trip
Flexible Fallback	Main Balance can cover spending if daily budget is exhausted
Bank Transfers	Users can transfer from Main Balance to any external Nigerian bank account via the Transfer Service

1.4 Target Users

- University students (primary)
- NYSC members
- Corporate workers
- Budget-conscious individuals

2. Security & Compliance

Control	Implementation
Authentication	JWT access tokens + refresh token rotation; PIN required for all transfer and high-value transaction approval
Idempotency	All financial operations carry an IdempotencyKey to prevent double-processing
Authorisation	Role-based (User / Vendor / Admin); wallet and transfer endpoints verify ownership before any operation
Webhook Verification	Paystack HMAC signature verified for deposit webhooks (Payment Service) and transfer webhooks (Transfer Service)
Audit Trail	WalletLedger records before/after balances and debit type for every money movement
Account Lockout	FailedLoginAttempts tracked; account locked after threshold exceeded
Transfer PIN Guard	All external bank transfers require PIN verification at Gateway before Transfer Service is called

KYC	Basic identity verification (MVP optional — flag for post-MVP)
Fraud Detection	Post-MVP phase — flag unusual spending or transfer patterns via analytics

3. Key Risks & Mitigations

Risk	Impact	Mitigation
User feels restricted by daily limits	Churn / poor reviews	Main Balance fallback; clear UX messaging about remaining daily budget
Duplicate transactions / double debit	Financial loss / data corruption	Idempotency keys enforced on every financial operation; Wallet checks key before any ledger write
In-platform payment race condition	Payer over-debited or recipient under-credited	Wallet atomic pay operation uses pessimistic DB lock; debit and credit in single transaction scope
Kafka consumer lag	Delayed balance updates visible to user	Monitor consumer group lag in Application Insights; Wallet balance read is sync REST — not event-dependent
Paystack deposit webhook failure	Missed deposits / user thinks money is lost	Manual verify endpoint in Payment Service + Paystack dashboard reconciliation; user can trigger re-check
Budget over-spend race condition	Budget balance goes negative	Pessimistic DB lock on Wallet debit; balance re-checked inside lock before commit
Transfer webhook failure	Funds pre-debited but transfer stuck Pending	Manual verify endpoint in Transfer Service; scheduled status poll every 30 min; auto-reversal after 24h timeout
Budget Service reacts to non-budget debits	Corrupt spend tracking	Budget Service subscribes only to wallet.budget.debited — never sees wallet.main.debited

4. MVP Scope

In Scope

- User registration, login, PIN setup, email verification
- Wallet: deposit via Paystack virtual account, Main + Budget balances, full ledger history
- Budget: create, daily release/expiry, FIFO spend tracking, gift budgets between users
- In-platform payments: vendor QR / UserID requests and user-to-user sends — purely internal wallet moves
- External bank transfers from Main Balance via Transfer Service (Paystack Transfers API)
- Transaction history with filtering by type, date, and status
- Push, email, and SMS notifications for all key events

Out of Scope (Post-MVP)

- Full KYC verification
- Fraud detection & ML anomaly scoring
- Analytics dashboard
- Multi-currency support
- Scheduled / recurring external transfers